Name of the Program: B.Sc. in CSE (94) **18 June 2021**
Semester: Winter 2020-2021 **Time: 2:30 PM – 4:00 PM**

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID-SEMESTER EXAMINATION **Winter Semester: 2020-2021**
Course Code: CSE 4303 **Full Marks: 75**
Course Title: Data Structures **Time: 1 Hour 30 Minutes**

There are **03 (three)** questions. Answer all of them. Figures in the right margin indicate marks. The examination is **Online** and **Closed Book.** Marks of each question and corresponding **CO** and **PO** are in the brackets. Renaming format 'fullID_cse4303.pdf'.

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1. a) The traffic on the Dhaka-Mymensingh highway is so unbearable nowadays that the government of Bangladesh has decided to build a Helipad in Gazipur so that VIP personnel can save time. Since Gazipur is the centre of many industries, this helipad caught the attention of many people and its usage is now in high demand. 25 (C01,C02, C03) (PO1,PO2, PO3,PO12)

Unfortunately, at present, there is only one helipad. To handle this high demand, the govt. has decided to launch a system named '**Helipad Reservation System (HRS)**' where reservation can be taken for any helicopter. The system will offer the following features:

i. ***Make_Reservation*** **(**int timeslot**):** The system only allows a reservation if the helipad is free within the three minutes range of the desired timeslot. For example, if a user wants to make a reservation in the kth minute, it will only be allowed if there is no reservation in the $(k-1, k-2, k-3, k+1, k+2, k+3)th$ minutes.

ii. ***Before_After*** **(**int timeslot**):** Once a reservation is confirmed, a query can be made to know the timeslot of the two reservations which are to be immediately served before & after the user's helicopter.

iii. ***Count_Before*** **(**int timeslot**)**: HRS also allows its users to make a query to know the number of reservations before any particular timeslot.

For example, if there are reservations at timeslots $\{25, 29, 50, 60, 75\}$ and user want to make a reservation at $timeslot = 45$, it will be added. After that, if there is another reservation-request at $timeslot = 42$, it won't be allowed as it falls within the $(k-3 \ldots k+3)$ range.

At present, the reservations are at {25,29,45,50,60,75}. So if the user makes the '*Before-After*(45)' query, the result will be 29 & 50. The results of '*Count-Before* (45)' query is 2, as there are two reservations (at $timeslot = 25$ & 29) before the timeslot 45.

Now your task is to propose a suitable way to store the reservations so that all of these requirements can be satisfied in minimal Time-complexity. Justify your choice of Data Structure and write appropriate pseudocodes to clarify how your choice will achieve these goals.

2. a) Evaluate the following Postfix expression using a stack (show detailed steps) — 8 (CO1) (PO1)

$$ID\ y\ 1\ +\ log\ \times\ a\ ID\ \div\ -$$

Note: 'ID' means the last two digits of your '$studentID$'. The characters 'y' & 'a' are considered as variables. Finally, $'log, +, -,\ \times, \div'$ are considered as operators.

b) For the set of keys $\{2,5,6,11,17,20, ID\}$, draw binary search trees of heights 2,3,4,5 and 6. Sequence of the numbers can be rearranged to adopt the heights. — 8 (CO1) (PO1)

Note: 'ID' means the last two digits of your '$studentID$'.

c) Explain the Space Complexity required for the Inorder-traversal of a binary tree. — 6 (CO1,CO3) (PO1,PO2,PO3)

d) Encode your 'Full-name' using Caesar-cipher. Use your '$studentID$' as the repeating key. — 3 (CO1), (PO1)

3. a) Suppose it is required to show the second-highest element from a set of numbers stored in a Max Heap. One of the ways to achieve this goal is to apply the '*Extract_Max*' operation twice. Criticize this approach and propose a better solution (if any). — 7 (CO1,CO3) (PO1,PO2,PO3)

b) Suppose an Array of size $'N'$ is being used to implement a $Circular\ Queue$. After a series of Enqueue/Dequeue operations, a state arrived where the queue is filled. In such states, it is required to increase the capacity of the queue. One idea is to use a new array of size $'2N'$ and directly copy the present state of the queue into the new array. Criticize this idea and propose a better one (if any). — 6+3 (CO1,CO3) (PO1,PO2,PO3)

How would the scenario be if $Linear\ Queue$ was used instead of $Circular\ Queue$?

c) Propose an algorithm to implement a 'First-in-First-out Queue' using a Priority Queue. — 9 (CO1,CO3) (PO1,PO2,PO3)